GOVERNMENT ENGINEERING COLLEGE, IDUKKI

(DEPT OF CSE)

ASSIGNMENT 1

PCCST302 THEORY OF COMPUTATION

Date of announcement: 21/07/2026 Max. Marks: 30
Date of submission: 03/08/2026 Faculty: Midhun Mathew

(Please specify your Group before answering)

GROUP A: FOR ROLL NUMBERS 1 to 20

1. Design DFA for { w / w is of even length and begins with 01 }
2. Design DFA for strings with exactly 3 consecutive zeros $\sum = \{0,1\}$
3. Design DFA for strings not containing the substring '110'
4. Design DFA for L = { w / w $\in$ {0,1}*, every zero in w has a one immediately to its right }
5. L = {$\in$, a, aa, aaa). $\sum = \{a\}$. Design NFA for L.

GROUP B: FOR ROLL NUMBERS 21 to 40

1. Design DFA for strings starting with 01 and ending with 10
2. Design DFA for strings in which number of a's and number of b's are multiples of 3
3. Design DFA for L = { $(01)^i (1)^{2j}$ / i, j $\geq$1}
4. Design DFA for strings exactly with two ones $\sum = \{0,1\}$
5. Design an NFA for binary numbers where first and last symbols are same.

GROUP C: FOR ROLL NUMBER 41 onwards

1. Design DFA for strings with neither consecutive a's nor consecutive b's
2. Design an NFA for strings with at least 2 a's. $\sum = \{a, b\}$
3. Design an NFA for strings in which first and last letters does not match. $\sum = \{a, b\}$
4. Design an NFA for L = { $\in$, 0, 1, 01, 10, 11}
5. Design a DFA for strings in which number of a's is a multiple of 2 and number of b's is a multiple of 3.